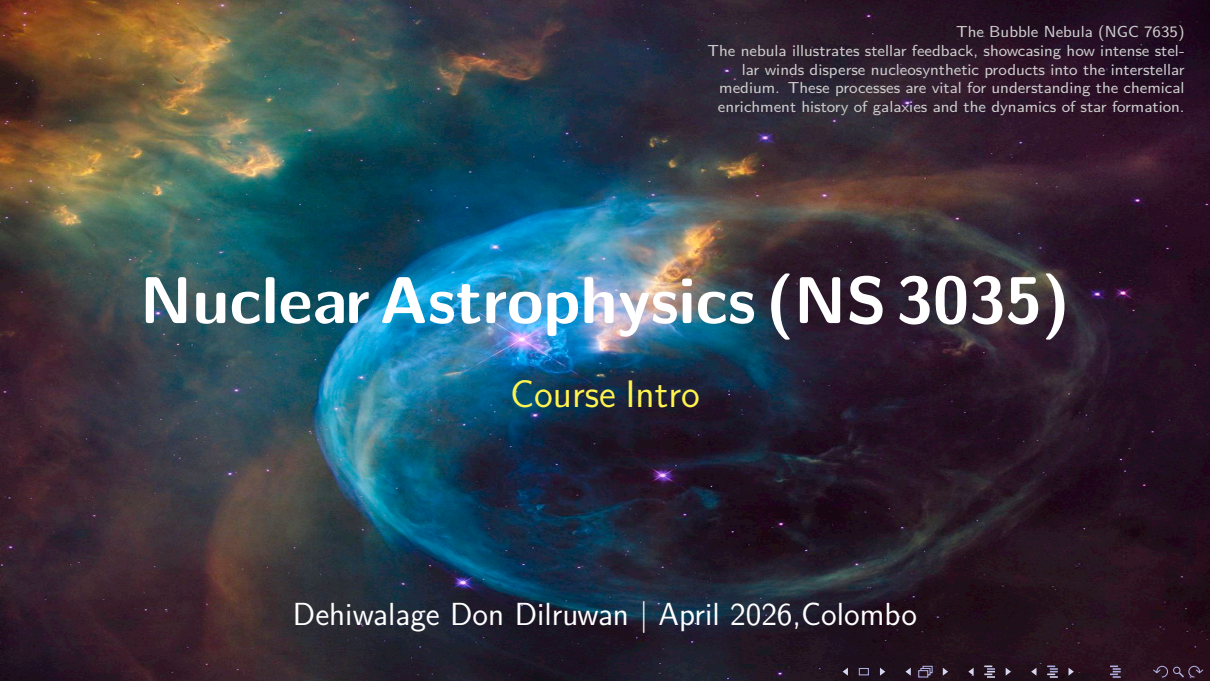




The Bubble Nebula (NGC 7635)

The nebula illustrates stellar feedback, showcasing how intense stellar winds disperse nucleosynthetic products into the interstellar medium. These processes are vital for understanding the chemical enrichment history of galaxies and the dynamics of star formation.

Nuclear Astrophysics (NS 3035)

Course Intro

Dehiwalage Don Dilruwan | April 2026, Colombo

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Introduction to Nuclear Astrophysics

Outline

Logistics

Introduction to Nuclear Astrophysics

Tentative Schedule

Module	Week	Primary Topic
SN	1	Course Plan & Intro to Astro
	2	Stellar Evolution & Compact Objects
	3	Thermonuclear Reactions in Stars
	4	Hydrostatic Burning I (H to He)
	5	Hydrostatic Burning II (Advanced)
EN	6	Supernova Mechanisms
	7	Thermonuclear & Exotic Events
	8	Nucleosynthesis Above Fe-peak
BBN	9	The Early Universe & BBN
Special	10	Lab, ML Methods & Projects

Project Guidelines and Topics

Evaluation: 50% of course grade

Groups: 2 students per group (5 groups total)

Credits: Final exam (1) and Project (1)

Criteria: report (4 pages + group presentation)

1. Bibliographic Study

Select state of the art paper

Summarize objectives/concepts

Extended bibliography review

Analyze

methodology/theory/data

Bonus: Critical analysis

Bonus: Reproduce results

2. Computational/Data

Select only if you are familiar with python, Linux

Use Python or MESA (1D stellar) and carry out

Nuclear Astrophysics project

Like ML or Data Science (I can help) ?

Requires clear data/objectives github repo/small

web site/ interactive visulization (3D blender)

Deadline: 2 weeks for proposal/paper selection.

Project Guidelines and Topics

Groups ?

Part 1: **Introduction to Nuclear Astrophysics**

Multidisciplinary Perspective

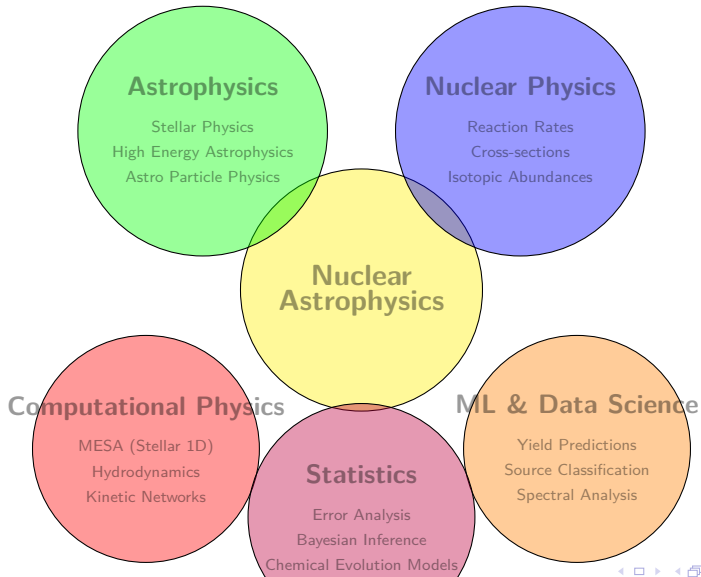


Outline

Logistics

Introduction to Nuclear Astrophysics

Nuclear Astrophysics is a Multidisciplinary Science



What is Astrophysics?

Astrophysics is the study of the Universe and its components

It evolved from astronomy towards astrophysics in the 1940-50's thanks to the discovery of

Nuclear Fusion: Powering stars (Eddington 1920; Payne 1925)

Expansion: Discovery of the expanding Universe (Hubble 1929)

Spectral Lines: Wollaston & Fraunhofer (18th c. breakthrough)

Let's talk about Cecilia Payne

The Discovery (1925)

Before Payne, many scientists assumed the Sun and stars had a composition similar to Earth

Cecilia Payne explains stars are mostly Hydrogen/Helium

First to apply quantum physics to astrophysics

"The most brilliant PhD thesis ever written in astronomy."



Cecilia Payne-Gaposchkin (1900–1979)

What Cecilia Payne Can Teach Us

Trust the Data: Truth often contradicts the common knowledge of the era.

Bridge Disciplines: Payne study botony. Attended general relativity discussions, and transitioned into astrophysics ... Just like that

Ignore Silos: Science is not limited by departments or institutional boundaries... progress depends on curiosity and persistence rather than prestige

Rejected from Cambridge for a degree solely because she was a woman

Institutional discrimination in academia should never stop scientific inquiry.



Cecilia Payne-Gaposchkin (1900–1979)

Astrophysics is a Large Field

Topics in Astrophysics include

Cosmology

Planetary sciences

Stellar physics

Asterosismology

Interstellar medium

Astroparticle physics

High energy astrophysics

Astrophysical plasmas

Instrumentation

Multimessenger astronomy

Note: Bold items denote core areas of *Nuclear Astrophysics* research.

What is Nuclear Physics?

Nuclear physics is the field of physics that study the structure, properties, and interactions of atomic nuclei.

Modern nuclear physics emerged from a series of experimental and theoretical breakthroughs such as

- Discovery of the Nucleus Ernest Rutherford (1911).

- Liquid Drop & Shell Models (Gammow, Maria)

- Quantum Chromodynamics (QCD)

Let's talk about Maria Goeppert Mayer

The Magic Numbers (1948)

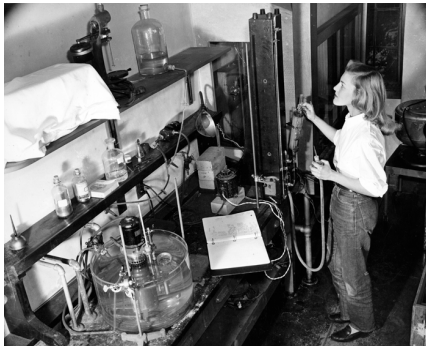
Proposed the **Nuclear Shell Model**.

Identified "Magic Numbers" (2, 8, 20, 28, 50, 82, 126) for nuclear stability

Revolutionized our understanding of nuclear architecture

Second women in history to win nobel price (after Madame Curie)

"Don't Grow up to be a women! Grow up to be a scholar"



Maria Goeppert Mayer (1906–1972)

1963 Nobel Prize in Physics

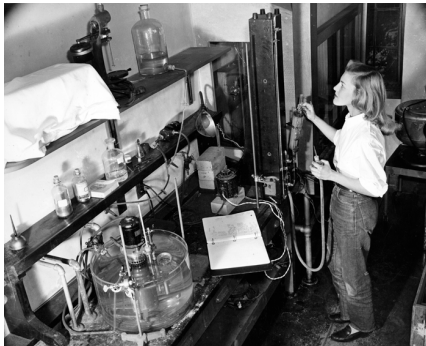
Lessons from Maria Goeppert Mayer

Question the Consensus: The "Liquid Drop" model was the standard, but Maria looked for the exceptions that broke the rule

Stay Curious: She worked in unpaid/volunteer positions for years, driven purely by the need to understand the physics

"S.D. Mother wins Nobel Prize" – 1925

Perseverance: She ignored administrative labels to focus on scientific reality



Maria Goeppert Mayer (1906–1972)

Nuclear Physics is a Large Field

Topics in Nuclear Physics include

Nuclear Structure: Ground-state properties, exotic nuclei

Nuclear Reactions: Fission, fusion, and scattering

Hadronic Physics: Quark-Gluon Plasma (QGP)

Nuclear Astrophysics: Areas where Nuclear Physics provides the fundamental "rules" for astrophysical phenomena

Applied Physics: Medical imaging, power generation, security

Fundamental Symmetries: Parity violation, neutrinoless double beta decay

Nuclear Astrophysics is Multidisciplinary Field

Definition: Nuclear Astrophysics is the interdisciplinary field that applies the laws of nuclear physics to explain the origin of elements, the evolution of stars, and the energetic processes of the cosmos

The Synthesis

Nuclear Physics (Micro) → *Strong force, reaction rates, binding energy*



Astrophysics (Macro) → *Stellar nucleosynthesis, galactic chemical evolution, compact objects*

Energy Sources: Fusion cycles powering stellar lifetimes

Nucleosynthesis: Origin of elements (Big Bang to Supernovae)

Dense Matter: Equation of State (EoS) of neutron stars

Cosmology: Primordial abundance and the early Universe

Asking the Right Questions



Master Picasso



Guernica (1937), Pablo Picasso

“Computers are useless. They can only give you answers. What is needed are questions!” —

Picasso

Why Guernica?

Picasso does not provide a simple answer.

He constructs a question about violence, suffering, and perception

Meaning emerges from interpretation not direct explanation

In science, **data alone is not discovery** questions define understanding

The challenge is to extract structure from chaotic information, whether in art or in the universe

Key Questions You will ask during this course ?

- (i) Where do the elements in the Universe come from?
- (ii) Are the atoms in your body the result of stellar death?
- (iii) How does nucleosynthesis convert light elements into heavy ones like Gold?
- (iv) Why do observations often conflict with our theoretical stellar models?
- (v) Why is the observed abundance of primordial Lithium lower than Big Bang predictions?
- (vi) What astrophysical engines power cosmic rays to ultra-relativistic energies?
- (vii) How does the debris from dying stars seed the next generation of star formation?